DBMS Theory Assignment :2

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B. Tech CSE Core

Section-A

Q (1) The Entity Relationship Model visually represents how entities relate in a database, offensive a clear blueprint of the data's organisation structure.

(2) In the relation model, data is organised into tables with relationships b/w entities defined through foreign key constraints.

(3) The object-oriented model treats data as a collection of objects, each with unique attribute and model, mirroring real-world entities

(4) Integrity const. ensure the reliability of data by enforcing rules that must be adhered to maintaining the accuracy of information.

Section-B

Q Healthcare system, discuss data abstraction and simplify the interactions.

Data abstraction is the process of hiding complex database details and showing only relevant information to users.
In a healthcare system data abstraction helps doctors, nurses and staff easily access patient info without dealing with complex database structures.

E.g. → Nurses can update patient vitals and treatment details.

P.T.O.

Date:

Through abstraction-level

(i) Physical Level – shows how data is stored in the database
(ii) Logical Level – shows the structure of the database like table & r/n.
(iii) View Level – shows only the table data required by specified users. This makes the system easier to user and secure.

Q3 Benefits and challenges of data abstraction

Benefits :

a) Simplicity – Healthcare staff can handle requires data without understanding database complexity.
b) Security – Info. can be restricted to authorized users only.
c) Data Integrity – Maintains accuracy & consistency of patient data.
d) Efficiency – Faster access to patient records improves medical decision making.

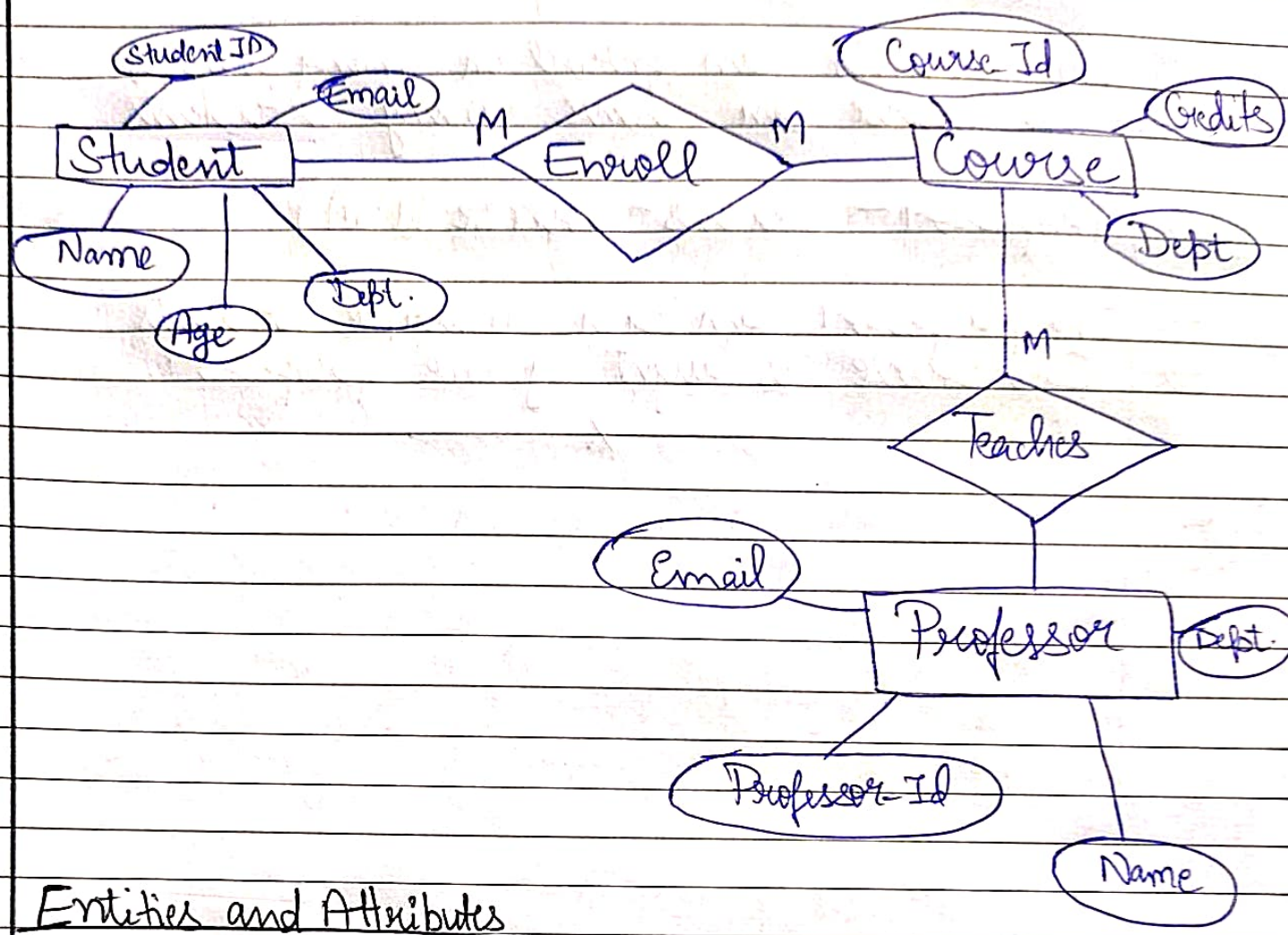
Challenges :

a) Implementation Complexity : Designing multiple abstraction level can be difficult.
b) Performance Overhead : Additional abstraction layers may slightly affect system performance.
c) Difficulty in maintenance

Section C

Q4 Design a database for university, with entities students, course and professor, using illustrations in ER. Also define the relationships between them.

Diagram:



Soln. Entities and Attributes

i) Student

Attributes

- Student ID (PK)
- Name
- Age
- Email
- Dept

(ii) Courses

Attributes

- Course ID
- Credits
- Dept.

(iii) Professor (Attributes)

- Professor_ID
- Email
- Department
- Name

Relationship -

(i) Student - Enrolls - Course (M:N)

→ One students can enroll in many courses
→ One course can have many students

(ii) Professor - Teaches - Course (1:M)

→ One professor can teach multiple students
→ Each course is taught by one professor.

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